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10 July 1969

# MILITARY SPECIFICATION

## INDICATORS, RATE OF FLOW, 0-5,000 PPH, INTEGRALLY LIGHTED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

* 1.1 *Scope.* This specification covers two types of remote indicating, integrally lighted, rate of flow indicators.

* 1.2 Classification. The indicators shall be of the following types, as specified (see 6.2):

| | |
|---|---|
| EFU-2/A | Integrally red lighted |
| EFU-12A/A | Integrally white lighted |

## 2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| QQ-P-416 | Plating, Cadmium (Electrodeposited) |
| PPP-B-636 | Boxes, Shipping, Fiberboard |

Military

| | |
|---|---|
| MIL-P-116 | Preservation-Packaging, Methods of |
| MIL-M-3171 | Magnesium Alloy, Processes for Corrosion Protection of |
| MIL-E-5400 | Electronic Equipment, Airborne, General Specification for |
| MIL-C-5541 | Chemical Conversion Coatings on Aluminum and Aluminum Alloys |
| MIL-S-7742 | Screw Threads, Standard, Optimum Selected Series: General Specification for |

FSC 6620

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| | |
|---|---|
| MIL-A-8625 | Anodic Coatings, For Aluminum and Aluminum Alloys |
| MIL-C-14806 | Coating, Reflection Reducing, for Instrument Cover Glasses and Lighting Wedges |
| MIL-S-20708 | Synchros, General Specification for |
| MIL-L-25467 | Lighting, Integral, Aircraft Instrument, General Specification for |
| MIL-T-26298 | Transmitter, Rate of Flow TRU-6/A, 0-5,000 PPH |
| MIL-L-27160 | Lighting, Instrument, Integral, White, General Specification for |
| MIL-B-27497 | Bearings, Jewel, Sapphire or Ruby, Synthetic |
| MIL-M-45202 | Magnesium Alloy, Anodic Treatment of |

Naval Air Systems Command

| | |
|---|---|
| SD-24 | General Specifications for Design and Construction of Aircraft Weapons Systems |

STANDARDS

Federal

| | |
|---|---|
| FED-STD-595 | Colors |

Military

| | |
|---|---|
| MIL-STD-100 | Engineering Drawing Practices |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-130 | Identification Marking of US Military Property |
| MIL-STD-143 | Standards and Specifications, Order of Precedence for the Selection of |
| MIL-STD-461 | Electromagnetic Interference Characteristics Requirements for Equipment |
| MIL-STD-462 | Electromagnetic Interference Characteristics, Measurement of |
| MIL-STD-701 | List of Standard Semiconductor Devices |
| MIL-STD-704 | Electric Power, Aircraft, Characteristics and Utilization of |
| MIL-STD-781 | Reliability Tests Exponential Distribution |
| MIL-STD-785 | Reliability Program For Systems and Equipment Development and Production |
| MIL-STD-794 | Parts and Equipment, Procedures for Packaging and Packing of |
| MIL-STD-810 | Environmental Test Methods |
| MIL-STD-831 | Test Reports, Preparation of |
| MIL-STD-889 | Dissimilar Metals |
| MS33639 | Case, Instrument, Clamp-Mounted, Aircraft |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)



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## 3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.4 and 6.4).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Preliminary list of electronic parts. A preliminary list of electronic parts shall be furnished directly to the procuring activity in accordance with MIL-E-5400 prior to ordering or fabricating parts for the production quantity. The parts shall conform to MIL-E-5400. Nonstandard parts shall be approved by the procuring activity prior to the fabrication of the completed item. Whenever practicable, nonstandard electronic parts shall not constitute more than 30 percent of the total quantity of electronic parts used in the construction of the indicator.

3.3 Materials

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the indicator, except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the indicator, except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

* 3.3.3.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Toxic and corrosive fumes. The indicator shall be of such material and construction that when the material is burned, fumes or gases that combine with the atmosphere to form acid or corrosive alkali will not be liberated.

3.3.5 Protective treatment. When materials are used in the construction of the indicator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.



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3.4 Design and construction

3.4.1 Operation. The indicator shall be designed to supply power to a TRU-6/A fuel-flow transmitter conforming to MIL-T-26298 and shall indicate the rate of flow in pounds per hour (pph) when connected as shown on figure 1. The indicator shall incorporate a synchro torque receiver in accordance with MIL-S-20708. Rotation of the indicator pointer shall be clockwise for an increasing flow rate.

3.4.2 Calibration. The indicator dial shall be graduated in accordance with figure 2. The calibration of fuel flow versus angular degrees shall be linear within $0^{o}$ ±15 minutes of arc within the flow range of 500 to 5,000 pph.

* 3.4.3 Integral lighting. The EFU-2/A indicator shall be designed for integral lighting in accordance with MIL-L-25467, and the EFU-12A/A indicator shall be designed for integral lighting in accordance with MIL-L-27160.

* 3.4.4 Compatibility. Both the EFU-2/A and EFU-12A/A indicators shall be completely compatible with TRU-6/A transmitters conforming to MIL-T-26298.

3.4.5 Transistors and diodes. Transistors and diodes used in the construction of the indicator shall be of the silicon type designed for continuous operation in a $70^{o}$C ($158^{o}$F) ambient temperature in accordance with MIL-STD-701. Circuit design shall be such that no electronic component will be operated outside its rated characteristics with 115V ±10 percent 400 Hz ±10 percent power input and with indicator input signals as specified herein. No electronic device shall break down when excited by transients as specified in MIL-STD-704.

* 3.4.6 Maintainability. The indicator shall be designed to provide ease of maintainability which shall include, but not be limited to, the following design requirements:

a. Minimization of the complexity of maintenance tasks (for example: calibratio adjustments, inspection, et cetera) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective parts, assembly, or component

d. Minimum numbers and types of tools and test equipment (special and standard) needed to perform maintenance

e. Optimum accessibility in all components requiring maintenance inspections, removals, or replacements.



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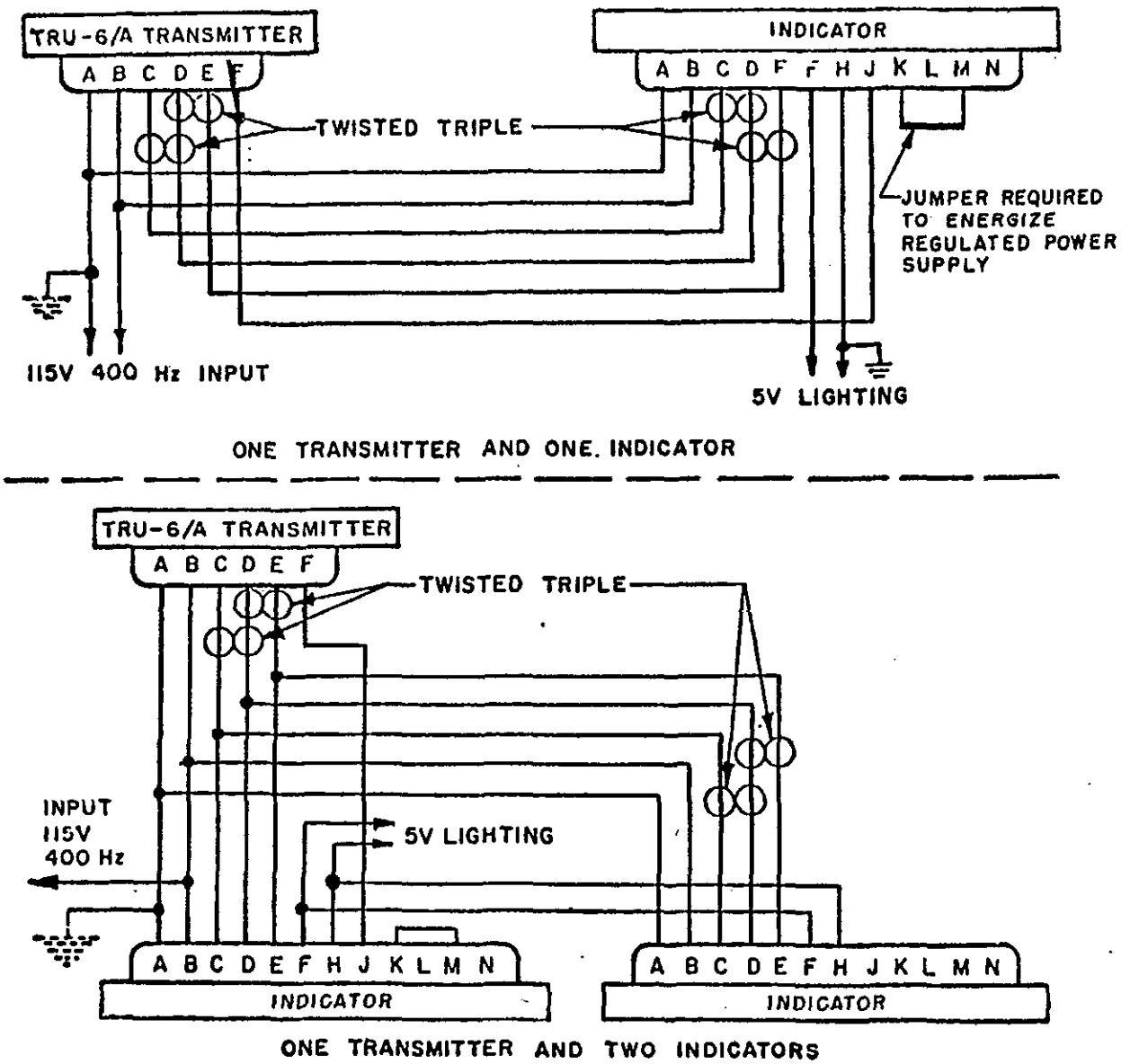



FIGURE 1. System Connections



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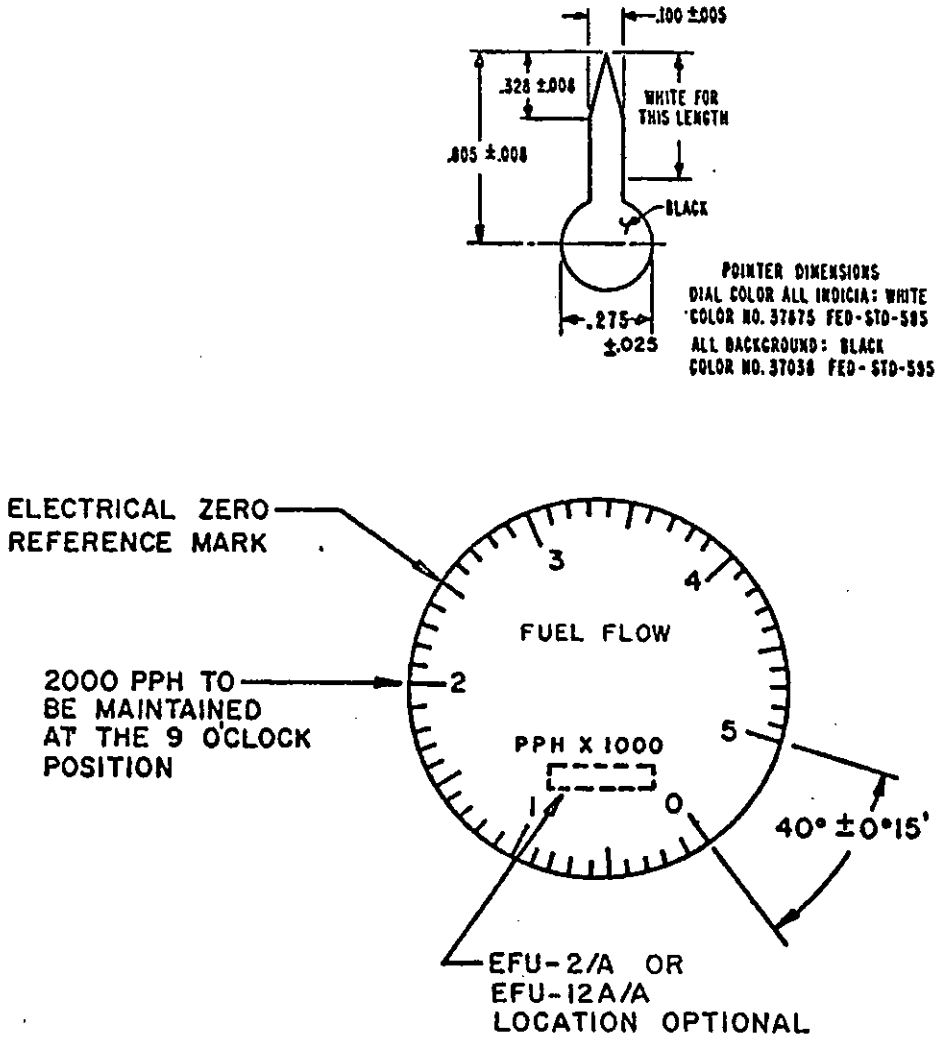



FIGURE 2. Dial and Pointer



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3.4.7 Reliability program. The contractor shall establish a reliability program and submit reports thereon in accordance with MIL-STD-785.

3.4.7.1 Mean-time-between-failures. The indicator shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,000 hours demonstrated to a 90 percent confidence level.

3.4.8 Longevity. The indicator shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 1,500 hours before wearout failures occur or the indicator consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

* 3.4.9 Power. The indicator shall operate within the tolerances specified herein when supplied with 115V, 400-Hz, single-phase power in accordance with category B of MIL-STD-704, except that the normal steady state limits of voltage and frequency may each vary ±10 percent. Operation with voltage surges to 195 ±5V shall be as specified in 4.6.13.

3.4.10 Pointer. The pointer shall be as shown on figure 2.

3.4.11 Dial. The dial shall have a minimum diameter of 1.750 inches as measured at the outside ends of the scale graduations. The dial shall be marked as shown on figure 2.

3.4.11.1 Electrical zero reference mark. The dial shall be provided with an electrical zero reference mark equidistant between the first and last graduations of the dial as shown on figure 2. The electrical zero reference mark shall be finished in durable dull white and shall be approximately 0.031 inch long, 0.010 inch wide, and shall be located on the outer edge of the dial so that it is visible through the cover glass.

* 3.4.11.2 Dial markings. The dial markings shall be finished in lusterless white, color No. 37875 of FED-STD-595, and shall conform to the following dimensions:

| | Height or length (inch ±0.016) | Width of line or graduation (inch +0.005) |
|---|---|---|
| Scale numerals 0 through 5 | 0.156 | 0.031 |
| 1,000-pph graduations | 0.156 | 0.031 |
| 500-pph graduations | 0.125 | 0.020 |
| 100-pph graduations | 0.094 | 0.020 |
| Lettering PPH X 1000 | 0.094 | 0.016 |
| Lettering FUEL FLOW | 0.125 | 0.020 |



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The marking EFU-2/A or EFU-12A/A, as applicable, shall be permanently and legibly marked on the dial in 0.047-inch letters. This marking and all other markings not otherwise specified shall be finished in lusterless black, color No. 37038 of FED-STD-595.

3.4.11.3 Visibility of dial. The pointer, numerals, at least 1/32 inch of the shortest graduations, and all other specified markings on the dial shall be visible from any point within the frustum of a cone the sides of which make an angle of 30° with a perpendicular to the dial and of which the small diameter is the aperture of the case.

3.4.12 Electrical zero. For the purpose of this specification, the indicator synchro rotor leads shall be designated $C_2$ and $H_2$ and the stator leads shall be designated $X_2$, $Y_2$, and $Z_2$. The subscripts indicate a requirement for low null at electrical zero. Identification of the leads and the determination of electrical zero shall be as follows: Arbitrarily label one rotor lead $C_2$, the low potential or ground lead, and label the other rotor lead $H_2$, the high potential lead. Arbitrarily label one stator lead $Z_2$ and connect $Z_2$ and $C_2$. Apply the rated voltage of the rotor between $C_2$ and $H_2$. Rotate the rotor to obtain the following conditions which indicate electrical zero, and label the remaining stator leads accordingly:

a. The voltage between $H_2$ and one of the unlabeled stator leads will be a maximum and the voltage between the unlabeled stator leads will be a minimum or at null.

b. The voltage between one of the unlabeled stator leads and the $Z_2$ lead shall increase in phase with the excitation voltage $C_2$ and $H_2$ before decreasing when the rotor is rotated in a clockwise direction from electrical zero as viewed from the shaft end. Label this lead $X_2$. Label the remaining stator lead $Y_2$. At electrical zero, the voltages $X_2$ to $Z_2$ and $Y_2$ to $Z_2$ shall be approximately in phase with excitation voltage $C_2$ and $H_2$. Schematically, the stator leads shall be designated $X_2$, $Y_2$, and $Z_2$ in a counterclockwise direction.

3.4.13 Case. The case shall conform to MS33639 as specified for the 2-inch size and shall be hermetically sealed. The seal shall be so designed and constructed that it will not be dependent upon materials that might be affected by the action of any atmospheric conditions to which the case may be subjected. The case length, dimension L, shall not exceed 6.125 inches, exclusive of the electrical receptacle. The design of the case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity.



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* 3.4.13.1 Case finish. The bezel and that part of the case visible from the cockpit side of the instrument panel shall be finished in lusterless black, color No. 37038 of FED-STD-595. The remainder of the case may be finished in black and shall be adequately protected to satisfactorily withstand the environmental conditions specified herein.

3.4.13.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter, dewpoint -65°C (-85°F), at the filling pressure. The filling medium shall be a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium shall be 12 to 14.7 psi. Other concentrations of helium shall not be used, except with prior approval of the procuring activity.

3.4.13.3 Filler tube. A tube or other facility for evacuating and filling the case shall be provided at the rear of the case and shall be recessed beneath the level of the back plate surface.

3.4.13.4 Cover glass. The cover glass shall be clear, flat, and free from flaws which would interfere with the normal reading of the indicator. The glass shall not be fluorescent when exposed to ultraviolet light or to rays in the 363 millimicron region. All reflecting surfaces shall be provided with a reflection reducing coating in accordance with MIL-C-14806.

3.4.14 Electrical receptacle. The indicator shall be provided with an electrical receptacle suitable for hermetic sealing similar to Cannon part number MC14H-13-12PN, or equal. The wiring of the receptacle shall be as shown on figure 3.

3.4.15 Synthetic jewel bearings. When sapphire or ruby synthetic jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.4.16 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

* 3.5 Performance. The indicator shall meet the requirements specified herein under the following conditions:

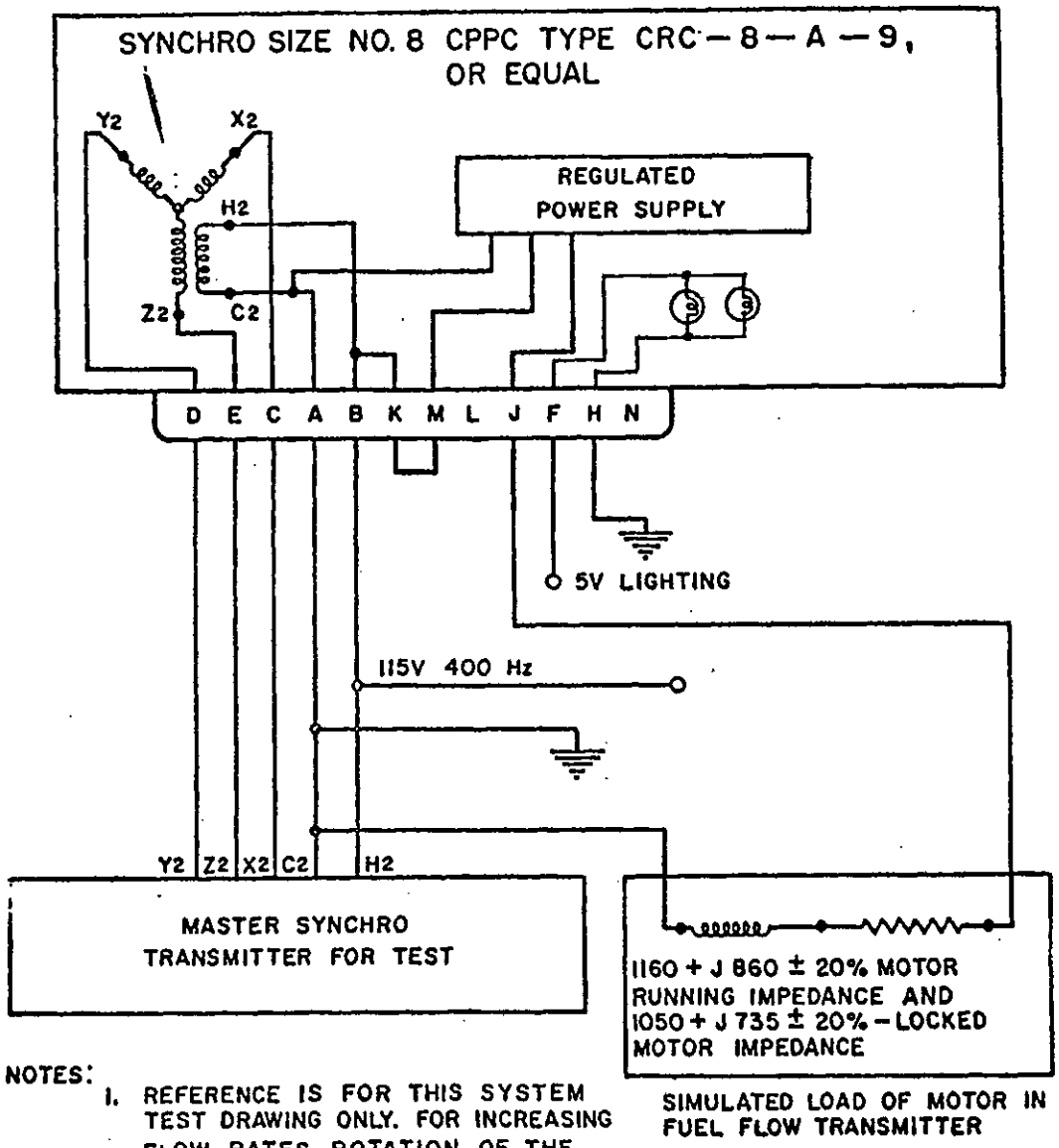
a. Operating temperatures ranging from -54°C to +71°C (-65° to +160°F) and storage temperatures ranging from -62°C (-80°F) to +80°C (+176°F)

b. Vibration from 5 to 500 Hz at double amplitude of 0.00012 to 0.010 inch for 9 hours

c. Eight cycles of alternate immersions in tap water at 85° and 5°C (185° and 41°F) for 30 minutes



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NOTES:

1. REFERENCE IS FOR THIS SYSTEM TEST DRAWING ONLY. FOR INCREASING FLOW RATES, ROTATION OF THE SYNCHRO ROTORS, AS VIEWED FROM THE SHAFT END, IS CLOCKWISE

FIGURE 3. Test Wiring



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d. Salt-fog for a period of 48 hours

e. Voltage and frequency variation ranging from 104V 296 Hz to 126V 484 Hz

f. Absolute pressure equivalent to 100,000 feet

g. Shock up to 15g

h. Frequency cycling from 300 to 500 Hz

i. Voltage surges to 195V

3.5.1 Fogging. There shall be no fogging of the face of the indicator glass when the indicator is energized and kept in a 70° ±2°C (158° ±4°F) controlled ambient temperature for a minimum of 1 hour and the glass face then rubbed with an ice cube for a period of 1 to 2 minutes.

* 3.5.2 Electromagnetic interference. The indicator shall meet the electromagnetic interference requirements of MIL-STD-461 as specified for class 1D equipment.

3.5.3 Magnetic effect. The indicators, operating or not operating, shall not create a magnetic field with an intensity that will deflect a short bar magnetic compass more than 3°.

.5.4 Response. The indicator shall show a rate indication from zero to half scale or from half scale to zero in less than 2-1/2 seconds under room- and high-temperature conditions.

3.5.5 Accuracy. The accuracy of the indicator shall be within the tolerances specified in tables I and II.

3.5.6 Dielectric strength. The indicator shall withstand applications of 500V at a commercial frequency applied between all pins not used for grounding and the case for a period of 5 seconds before the indicator is sealed, and 200V after the indicator is sealed.

3.5.7 Power consumption. With 115 ±1V 400 ±2 Hz power applied and with the internal regulated power supply de-energized, the power consumption shall be a maximum of 8.0 va and shall not exceed 24 va with the regulated power supply energized and supplying power to a 1,160 +J860-ohm load.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.



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* TABLE I. Test Tolerances

| Test Requirements | Tolerance, ±pph |
|---|---|
| Pointer indication tests: | |
| Position | 10 |
| Voltage and frequency variation | 1-1/2 percent of test point above 500 pph |
| Vibration | |
| Pointer oscillation double amplitude | 20 |
| Pointer variation | 20 |
| Friction: | |
| Room temperature | 10 |
| High temperature | 10 |
| Low temperature | 15 |
| Response time 0-5,000 pph (The pointer shall not overshoot the final position more than one time.) | 5 sec (max) 2 sec (min) |
| Regulated power supply output: | |
| Room temperature | 70-126V 400 ±1 Hz |
| High temperature | 70-126V 400 ±1 Hz |
| Low temperature | 70-126V 400 ±1 Hz |
| Voltage and frequency variation | 70-126V 400 ±1 Hz |
| Vibration | 70-126V 400 ±1 Hz |

3.7 Weight. The weight of the indicator shall not exceed 1.50 pounds.

3.8 Finishes and protective coatings

3.8.1 Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

a. Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541

b. Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing

c. Castings containing nonaluminum alloy inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing



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d. Small holes and inserts and small areas that need to be machined after the part is anodized need not be anodized

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

TABLE II. Indicator Fuel Flow Rate Tolerances

| Fuel Flow PPH | Tolerance | |
|---|---|---|
| | Room Temperature ±PPH | High and Low Temperature ±PPH |
| 0 | 15 | 25 |
| 200 | 15 | 25 |
| 500 | 15 | 25 |
| 1,000 | 15 | 25 |
| 1,500 | 10 | 25 |
| 2,000 | 10 | 20 |
| 2,500 | 10 | 25 |
| 3,000 | 15 | 25 |
| 3,500 | 15 | 25 |
| 4,000 | 15 | 25 |
| 4,500 | 20 | 30 |
| 5,000 | 20 | 30 |

3.8.2 Magnesium alloy parts. Magnesium alloy parts shall not be used in the indicator, except when essential. When essential, for Air Force use such parts shall be covered with an anodic film of the HAE type conforming to MIL-M-45202. When abrasion resistance is a factor, film in accordance with MIL-M-3171 shall not be used in lieu of the anodic-type HAE film. For the Navy, the use of magnesium shall conform to SD-24.

3.8.3 Where practicable, steel parts shall be cadmium plated in accordance with type II or III of QQ-P-416, and shall be of a class that is adequate to achieve the degree of protection required.

3.8.4 Finishes and protective coatings that will crack, chip, or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.9 Electrical schematic. An electrical schematic showing the function of each connector pin shall be permanently attached to the indicator. An approved metal decalcomania may be used provided it is permanently attached and cannot be peeled back with the fingernail.



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3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10.1 Nameplate. A nameplate shall be provided on the rear of the case. When applicable, the information shall conform to MIL-STD-130. Metal decal-comanias shall not be used.

3.11 Workmanship. The indicator shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antiblacklash techniques shall be used.

## 4. *QUALITY ASSURANCE PROVISIONS*

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of indicators shall be classified as follows:

a. Qualification tests.....See 4.4

b. Quality conformance tests.....See 4.5

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature, approximately 25°C (77°F). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

*4.3.2 Test position. Unless otherwise specified (see 6.2) the indicator shall be tested with the case in a horizontal position with the scale in the normal viewing position.



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* 4.3.3 Master transmitter. The indicator shall be tested with a master synchro transmitter having an error of ±2pph maximum that generates a signal equivalent to the signal output of a TRU-6/A rate of flow transmitter in accordance with MIL-T-26298.

4.3.4 Tolerances. Test tolerances shall be as specified in tables I and II.

4.3.5 Test power. Unless otherwise specified (see 6.2), the indicator shall be tested with 115 ±3V, 400 ±2 Hz single-phase power, and 5.0 ±0.1V ac or dc power for lighting.

4.3.6 Test load. Unless otherwise specified (see 6.2), the indicator shall be tested while connected to a transmitter or simulated transmitter load. The test wiring that may be used is shown on figure 1 or 3, as applicable. Unless otherwise specified, if a simulated transmitter load is used it shall be applied initially simulated as a locked motor load and then changed to a simulated running motor load. The simulated locked motor load shall be applied for 1 or 2 seconds if the indicator temperature is 10°C (50°F) or higher or for 3 to 5 seconds if the indicator temperature is below 10°C (50°F).

4.3.7 Lighting. Unless otherwise specified herein, indicator lighting shall be energized and shall operate satisfactorily when the indicator is energized and operating during testing.

4.4 Qualification testing

4.4.1 Test samples. The test samples shall consist of 14 indicators representative of the production equipment. Eleven of the samples shall be subjected to the individual, reliability, and longevity tests only and the other three samples shall be subjected to all the tests except reliability and longevity. The samples shall be identified with the manufacturer's part number and such other information as required by the qualifying activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831.

4.4.3 Qualification tests. The qualification tests shall consist of all the tests specified under 4.6. Unless otherwise specified (see 6.2), the tests shall be conducted in the order listed.

4.5 Quality conformance tests. The quality conformance tests shall consist of the following:

a. Individual tests.....See 4.5.1

b. Sampling plans and tests....See 4.5.2.



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* 4.5.1 Individual tests. Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product.....See 4.6.1

b. Electrical zero test.....See 4.6.2

c. Dielectric strength test.....See 4.6.3

d. Scale error test.....See 4.6.4

e. Friction test.....See 4.6.5

f. Response test.....See 4.6.6

g. Sealing test.....See 4.6.7

h. Burn-in test.....See 4.6.8

i. Fogging test.....See 4.6.9

j. Lighting (individual tests of MIL-L-25467 or MIL-L-27160).

* 4.5.2 Sampling plans and tests

* 4.5.2.1 Sampling plan A. A minimum of three indicators shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Individual tests.....See 4.6.1

b. Vibration resonance search.....See 4.6.10

c. Voltage and frequency variation test..See 4.6.11

d. Frequency cycling test.....See 4.6.12

e. Voltage surge test.....See 4.6.13

f. Low temperature test.....See 4.6.14

g. High temperature test.....See 4.6.15

h. High altitude-low temperature test....See 4.6.16

i. Position error test.....See 4.6.17



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j. Power consumption test.....See 4.6.19

k. Lighting (sampling plan A tests of MIL-L-25467 or MIL-L-27160).

* 4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three indicators shall be selected at random from the first 10 items produced on each contract or order and subjected to the following tests in the order specified (unless otherwise approved by the procuring activity) and as described under 4.6. If any one of the samples specified in sampling plan A is selected for sampling plan B, the sampling plan A tests shall not be repeated on that sample.

a. Sampling plan A tests

b. Magnetic effect test.....See 4.6.18

c. Frequency cycling test.....See 4.6.12

d. Low temperature exposure test.....See 4.6.20

e. High temperature exposure test.....See 4.6.21

f. Salt-fog test.....See 4.6.22

g. Thermal shock test.....See 4.6.23

h. Shock test.....See 4.6.24

i. Electromagnetic interference test....See 4.6.25

j. Lighting (sampling plan B tests (including life test) of MIL-L-25467 or MIL-L-27160)

k. Vibration test.....See 4.6.26.

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, indicators still on hand or later produced shall not be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, the tests specified by the procuring activity shall be repeated.

4.5.2.3.1 Reconditioning of tested indicators. Unless otherwise specified (see 6.2), indicators that have been subjected to the sampling tests shall be reconditioned by the contractor by replacing all worn or damaged parts. After reworking, the contractor shall resubmit the indicators for acceptance.



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4.5.2.4 Individual tests may continue. Individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of indicators on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

* 4.5.2.5 Reliability sampling. For reliability sampling, a lot shall consist of all indicators produced on a specific contract. Unless otherwise specified by the procuring activity, a minimum of five indicators shall be selected at random from each lot of 100 or fraction thereof and subjected to the test specified in 4.6.29.2.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in indicators already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, dimensions, and weight.

4.6.2 Electrical zero test. For the purpose of this test, a single indicator with the glass removed shall be connected to the indicator being tested and used as a transmitter unit. The standard transmitter shall not be used. With the power source disconnected, the pointer of the single indicator shall be displaced 180° from the pointer position of the indicator being tested. The pointer of the single indicator shall be held in this position and power shall be applied. This procedure shall be repeated a minimum of three times after which electrical zero shall be imposed on the indicator being tested. The tip of the pointer shall be within 0.10 inch of the electrical zero reference mark.

4.6.3 Dielectric strength test. Prior to sealing, 500V at a commercial frequency shall be applied between certain pins and the case for a period of 5 seconds. Only those pins not used for grounding the components to the case shall be used. After sealing, 200V at a commercial frequency shall be applied between only those pins not used for grounding and the case for a period of 5 seconds. There shall be no insulation breakdown. All lighting pins shall be considered grounded for this test.

4.6.4 Scale error test. The indicator shall be connected to a master transmitter. The transmitter shall be set at angles equivalent to each of the test points in table II. The indicator shall read within the tolerances specified therein for the applicable range. The output of the indicator power supply shall also be within the tolerances specified in tables I and II.



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4.6.5 Friction test. The indicator shall be tested for friction at scale error test points 5, 10, 50, and 90 percent of full scale. The master transmitter shall be made to slowly approach the test point in an ascending direction. After the transmitter has been held at the test point for approximately 5 seconds, the indicator pointer reading shall be recorded. The indicator shall then be gently vibrated to establish a stable condition, and the reading shall be recorded. The difference between the two readings shall be recorded as friction and shall not exceed the tolerances specified in table I.

4.6.6 Response test. The indicator shall respond and indicate a half-scale or full-scale reading within 1 percent when a half-scale step input is applied from zero to half-scale or from half-scale to full-scale, respectively, within the response tolerances specified in table I. The indicator shall not be energized for at least 3 minutes prior to this test.

4.6.7 Sealing test. The indicator case shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of 1 atmosphere.

4.6.8 Burn-in test. The indicator shall be placed in a chamber and the temperature maintained at 50°C (122°F) for a minimum of 24 hours. Power shall be applied in accordance with 4.3.6 throughout the test while maintaining a constant signal input equivalent to a half-scale indication. At least 8 hours of this test shall be performed during normal working hours, and observations shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the indicator display or the power source voltage output change by more than one-half of the tolerance specified for the test point in the applicable test tolerance table. Pointer oscillation or erratic movement shall be cause for rejection. The indicator shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.4.

4.6.9 Fogging test. The indicator shall be energized and kept in a 70° ±2°C (158° ±4°F) controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at the high temperature, an ice cube shall be rubbed on the face of the indicator cover glass for a period of 1 to 2 minutes. The glass shall then be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of water and oil fog. Any evidence of fogging on the inner surface of the cover glass shall be cause for rejection. This test may be conducted in conjunction with the high temperature scale error and friction tests.

* 4.6.10 Vibration resonance search. The indicator shall be subjected to a vibration resonance search in accordance with MIL-STD-810, Method 514.1, procedure 1, Part I, equipment category (a), curve B. The time shall be limited to two 15-minute sweep cycles from 5 to 500 to 5 Hz. During this test, a signal input shall be applied to the indicator to provide an indication of approximately half-scale. The pointer oscillation and variation shall not exceed the tolerances specified in table I.



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4.6.11 Voltage and frequency variation test. This test shall be conducted with the indicator connected as shown on figure 3 and with a load impedance of 1,160 +J860 ohms, ±20 percent. Power shall be applied to obtain the following combinations of voltage and frequency: 104 ±1V 300 ±2 Hz; 104 ±1V 500 ±2 Hz; 126 ±1V 300 ±2 Hz; and 126 ±1V 500 ±2 Hz. With each combination of voltage and frequency, accuracy of the indications shall be determined by positioning the master transmitter to obtain simulated flow rates of 1,000, 2,000, and 4,000 pph. Accuracy of the indicated flow rates and the output of the regulated power supply shall be as specified in table I. Prior to this test, the error of the master transmitter shall be determined at the specified test points using the specified voltages and frequencies. The transmitter error shall be algebraically subtracted from indicated flow rates to determine the indicated error.

* 4.6.12 Frequency cycling test. With the indicator connected as specified in 4.3.7, the voltage shall be adjusted to 115 ±1V and the frequency cycled from 360 to 440 Hz in accordance with the following schedule with approximately half-scale flow:

a. Start cycle at 360 Hz, increase frequency to 440 Hz in 38 seconds linearly with time

b. Return frequency to 360 Hz in 2 seconds or less

c. Increase frequency to 440 Hz in 58 seconds linearly with time

d. Return frequency to 360 Hz in 2 seconds

This 100-second cycle shall be repeated continuously for 8 hours at the rate of 36 cycles per hour. Power shall then be removed for a period of 2 ±1/2 hours and shall be reapplied at 121 ±1V 320 ±2 Hz for 1 hour. Immediately following this 1-hour period, the indicator shall be frequency cycled for 1 hour in accordance with the schedule indicated in a,b, and c except the frequency range shall be increased to between 300 and 500 Hz. The indicator shall then be subjected to the scale error test at room temperature.

4.6.13 Voltage surge test. The indicator shall be connected as shown on figure 1 or 3, as applicable. This test shall be made at approximately 5, 50, and 90 percent of full scale. Rated power shall be applied for 30 minutes. The power input shall then be adjusted to 108 ±1V 480 ±2 Hz for 1 hour and adjusted immediately to 121 ±1V 480 ±2 Hz for 30 minutes. At the end of this 2-hour operating period, the power input shall be adjusted to 115 ±1V 320 ±2 Hz. A voltage surge shall then be applied to provide an rms voltage of 160 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.09 ±0.005 second. The frequency shall then be increased to 480 ±2 Hz and a voltage surge shall be applied to provide a momentary



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rms voltage of 195 ±5V and return to 115 ±1V. The duration of the surge, including rise and decay time, shall be 0.07 ±0.005 second. Each surge shall be applied 10 times. At the end of this test, the voltage and frequency shall be adjusted to 115 ±1V 400 ±2 Hz and a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of tables I and II.

4.6.14 Low temperature test. The indicator, not energized, shall be subjected to a temperature of -54° ±2°C (-65° ±4°F) for a period of 4 hours. While at this temperature, the master transmitter shall be set at zero pph. Power shall be applied to the indicator. The output of the power supply shall be monitored for compliance with table I. At the end of a 10-minute period, the master transmitter shall be set to the half-scale test point, and the time for the indicator to reach this point shall not exceed 25 seconds. The indicator shall then be subjected to the friction and scale error test and shall meet the tolerances specified in tables I and II. While the indicator is still in a -54°C (-65°F) ambient, the response test shall be conducted. The response time at -54°C (-65°F) shall not exceed 10 seconds. This test shall be conducted immediately after the low temperature scale error test and before power has been applied for 40 minutes.

* 4.6.15 High altitude-low temperature test. After the low temperature test and while the indicator is still at -54°C (-65°F) ambient, the indicator shall be subjected to an absolute pressure equivalent to an altitude of approximately 100,000 feet. While at this pressure and temperature, the indicator shall be subjected to a scale error test specified in 4.6.4. The scale error shall not exceed the tolerances specified in table II. There shall be no damage to the indicator as a result of this test.

4.6.16 High temperature test. The indicator, while energized, shall be subjected to a temperature of 71° ±2°C (160° ±4°F) for a period of 4 hours and tested at this temperature for friction and scale error specified in tables I and II. At the conclusion of the test and while the indicator is still in a 71°C (160°F) ambient, the response time at 71°C (160°F) shall not exceed 3 seconds. This test shall be conducted immediately after the high temperature scale error test. The output of the power supply shall be checked for compliance with table I.

* 4.6.17 Position error test. The scale error test shall be repeated at the scale error test points nearest to 5, 10, 50, and 90 percent of full scale with the instrument turned 90° right, 90° left, face up, and face down. The change in scale error shall not exceed the tolerances specified in table I.

4.6.18 Magnetic effect test. The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in



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positions 0°, 45°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated on its own horizontal axis until it is in the normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°. The test shall be repeated with the indicator operating at rated voltage.

4.6.19 Power consumption test. With 115 ±1V 400 ±2 Hz power applied and the regulated power supply de-energized, indicator power consumption shall be a maximum of 8.0 va, exclusive of the lighting circuit, and shall not exceed 24 va with the regulated power supply energized and supplying power to a 1,160 ±J860-ohm load.

4.6.20 Low temperature exposure test. The low temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 502, except that after the 48-hour exposure at -62°C (-80°F), the temperature shall be raised to -54°C (-65°F) and maintained for 24 hours. Upon completion of the 24-hour period at -54°C (-65°F) and while still at this temperature, the indicator shall be subjected to the scale error test and shall meet the tolerances specified in 4.6.14. The indicator shall then be maintained at room temperature for a period of not less than 4 hours and subjected to the scale error at room temperature test. The scale errors shall not exceed the tolerances specified. The indicator shall be examined for deterioration or damage due to extreme temperature exposure.

4.6.21 High temperature exposure test. The high temperature exposure test shall be conducted in accordance with MIL-STD-810, Method 501, procedure I except as specified herein. During the entire 48-hour period, the indicator shall be energized. At the end of the 48-hour period and while still at 71°C (160°F), the indicator shall be tested for scale errors which shall not exceed those specified in 4.6.15. The indicator shall then be maintained at room temperature for a minimum period of 4 hours and again subjected to the scale error test. The errors shall not exceed the specified tolerance. The indicator shall be examined for deterioration or damage due to the extreme high temperature exposure.

4.6.22 Salt-fog test. The indicator shall be subjected to a salt fog test in accordance with Method 509 of MIL-STD-810. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerance after this test shall not exceed the room temperature tolerances specified in table II. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including the electrical receptacle, shall be cause for rejection.

4.6.23 Thermal shock test. The indicator shall be subjected to a total of 8 cycles of immersion in tap water at a temperature of 85° ±2°C (185° ±4°F) and then in tap water at a temperature of 5° ±2°C (41° ±4°F). The length of time



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for each bath immersion shall be 30 minutes with not more than 3 seconds between immersions. The indicator shall then pass the test specified in 4.6.7.

4.6.24 Shock test. The indicator shall be subjected to a shock test in accordance with MIL-STD-810, Method 516.1, procedure I, except the shock force shall be 15g. Immediately after this test, the indicator shall be subjected to the scale error test at room temperature. The error shall not exceed the tolerance specified.

* 4.6.25 Electromagnetic interference test. The electromagnetic interference test shall be conducted in accordance with MIL-STD-462, except that the frequency range shall be from 0.15 to 150 mc. Test methods shall be as follows: CE02, CE03, CE04, CS01, CS02, CS06, RS01, RS02, and RS03.

* 4.6.26 Vibration test. The indicator shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514.1, procedure I, equipment category (a), curve B. The indicator shall be operating and shall be vibrated in each of the three mutually perpendicular planes with a signal input of approximately half scale. Following the vibration period, the indicator shall be subjected to the scale error and friction error tests. Room temperature tolerances shall be within those specified. The indicator shall be opened and examined and no screws or other parts shall show evidence of loosening or damage as a result of this test.

4.6.27 Lighting tests. The indicator shall be subjected to all the lighting tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.6.28 Compatibility tests. The indicator shall be subjected to compatibility tests to determine compliance with 3.4.1, 3.4.2, 3.4.4, 3.4.9, 3.4.11.1, 3.4.12, 3.5.4, 3.5.5, and 3.5.7.

4.6.29 Reliability

4.6.29.1 Qualification. The 11 indicators specified in 4.4.1 shall be subjected to a reliability test in accordance with MIL-STD-781, test level E, which shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level.

4.6.29.1.1 Test equipment. The test equipment and fixtures shall include provisions for indicator operation cycling, temperature cycling, and vibration.

4.6.29.1.2 Operation cycling. During this test, the indicator shall be continuously cycled from 5 to 90 percent to full scale and return at a rate of 1 cycle every 4 minutes.

4.6.29.1.3 Temperature cycling. Unless otherwise specified or amended by the results of the thermal analysis, the temperature of the chamber in which



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the indicators are tested shall be cycled as follows:

Step a. Starting at room temperature, decrease temperature to -54°C (-65°F) and stabilize for 1-1/2 hours with indicators unenergized.

Step b. Turn indicators on and raise temperature to 55°C (131°F) and operate indicators for 4 hours

Step c. Turn indicators off and lower chamber temperature to -54°C (-65°F) for 2 hours

Step d. Repeat steps b and c until reliability requirement has been met. This schedule provides four 6-hour test cycles for each 24-hour period. In each 24-hour period, the operation test time is 16 hours.

4.6.29.1.4 Vibration cycling test. Throughout this test, the indicators shall be vibrated for a period of 10 minutes during each hour of on-time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g ±10 percent.

4.6.29.1.5 Performance check. At each 100-hour interval and at the end of this test, the indicators shall be subjected to the following performance tests: Scale error, friction, response time, and lighting (color and brightness tests). During each scale error test, if any two test points exceed the allowable temperature tolerances specified in table I or II, as applicable, one failure shall be charged against this test. The lighting circuit shall be energized during the entire test. The indicators shall meet the lighting tolerances of the applicable lighting specification (MIL-L-25467 or MIL-L-27160).

4.6.29.1.6 Failure criteria. For an accept decision, each indicator shall be operated at least 600 hours. If an indicator fails, entries shall be made of the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the indicators still under test. The procuring activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I, after each indicator has been operated for at least 600 hours.

4.6.29.1.7 Reporting. Reports shall be submitted in accordance with MIL-STD-781. The reliability test log and the failure summary and analysis shall be included in the final test report.



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* 4.6.29.2 Production sampling. The sample indicators specified in 4.5.2.5 shall be subjected to a reliability test in accordance with MIL-STD-781, test level E, paragraph entitled Reduced Testing, and as specified in 4.6.29 which shall demonstrate a minimum acceptable MTBF of 1,000 hours to a 90 percent confidence level. Accept-reject criteria shall be in accordance with test plan V except the discrimination ratio shall be 1.5.

4.6.30 Longevity tests. Testing of 3 of the 11 reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, until 1,500 hours of operation have accumulated on each indicator.

* 4.6.31 Maintainability demonstration. It shall be demonstrated that the maintainability requirements of 3.4.6 have been satisfactorily met.

4.7 Inspection of preservation, packaging, packing, and marking for shipment and storage. The inspection of preservation, packaging, packing and marking for shipment and storage shall be in accordance with section 5.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

* 5.1.1 Level A. Indicators shall be preserved and packaged in accordance with MIL-STD-794 and method III of MIL-P-116, one each in a unit container conforming to PPP-B-636, weather-resistant class. Electrical receptacles and connectors shall be equipped with water-vaporproof caps.

* 5.1.2 Level C. Indicators shall be individually preserved and packaged in a manner that will provide protection against corrosion, deterioration, or physical damage so that serviceability is assured during shipment from supply source to the first receiving activity. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

* 5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed for consolidation in shipping boxes conforming to PPP-B-636, weather-resistant class. Exterior shipping containers shall be of uniform shape and size, of minimum tare, weight, and cube.

* 5.2.2 Level B. Indicators, preserved and packaged as specified in 5.1.1 shall be packed for consolidation in exterior shipping boxes conforming to PPP-B-636, domestic class. Other requirements as specified in 5.2.1 shall apply.



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* 5.2.3 Level C. Indicators shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery, and shall meet as a minimum, the requirements of carrier rules and regulations. This level may be the contractor's commercial practice provided the latter meets the criteria for this level as defined in MIL-STD-794.

* 5.3 Marking. In addition to the standard marking requirements of MIL-STD-129, level C/C packs containing item(s)/equipment intended for immediate use at destination shall be conspicuously marked as follows on two opposite sides of the exterior container in letters as large as space permits:

FOR IMMEDIATE USE
DO NOT STORE IN THIS PACK

5.3.1 Nomenclature. The nomenclature shall be as follows:

INDICATOR, RATE OF FLOW ___#___, 0-5,000 PPH,
2-INCH SIZE, INTEGRALLY LIGHTED

# Applicable designation (EFU-2/A or EFU-12A/A) to be entered by the contractor.

5.3.2 Reinspection date marking. The reinspection date marking shall be as specified by the procuring activity.

6. NOTES

* 6.1 Intended use. The indicators covered by this specification are intended for use in aircraft to indicate the fuel rate of flow in pounds per hour to an engine.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Type indicator required (see 1.2)

c. If test position is to be other than as specified in 4.3.2

d. When test power is to be other than as specified in 4.3.5



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e. When test load conditions are to be other than as specified in 4.3.6

f. When tests are not to be conducted in the sequence specified in this specification (see 4.4.3)

g. When sampling plan B tests are not to be conducted (see 4.5.2.2)

h. If indicators that have been subjected to the sampling tests are not to be reconditioned (see 4.5.2.3.1)

i. Required levels of preservation and packaging and packing (see section 5)

* 6.3 Data. Data generated by this specification is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this document is as follows:

a. A preliminary list of electronic parts (see 3.2.1)

b. Reliability program reports (see 3.4.7)

c. Qualification test report (see 4.4.2)

d. Detailed analysis as specified in 4.6.29.1.6

e. Reports as required by 4.6.29.1.7.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASD/ENFIP, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity.

* 6.5 Identification of changes. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to



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evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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